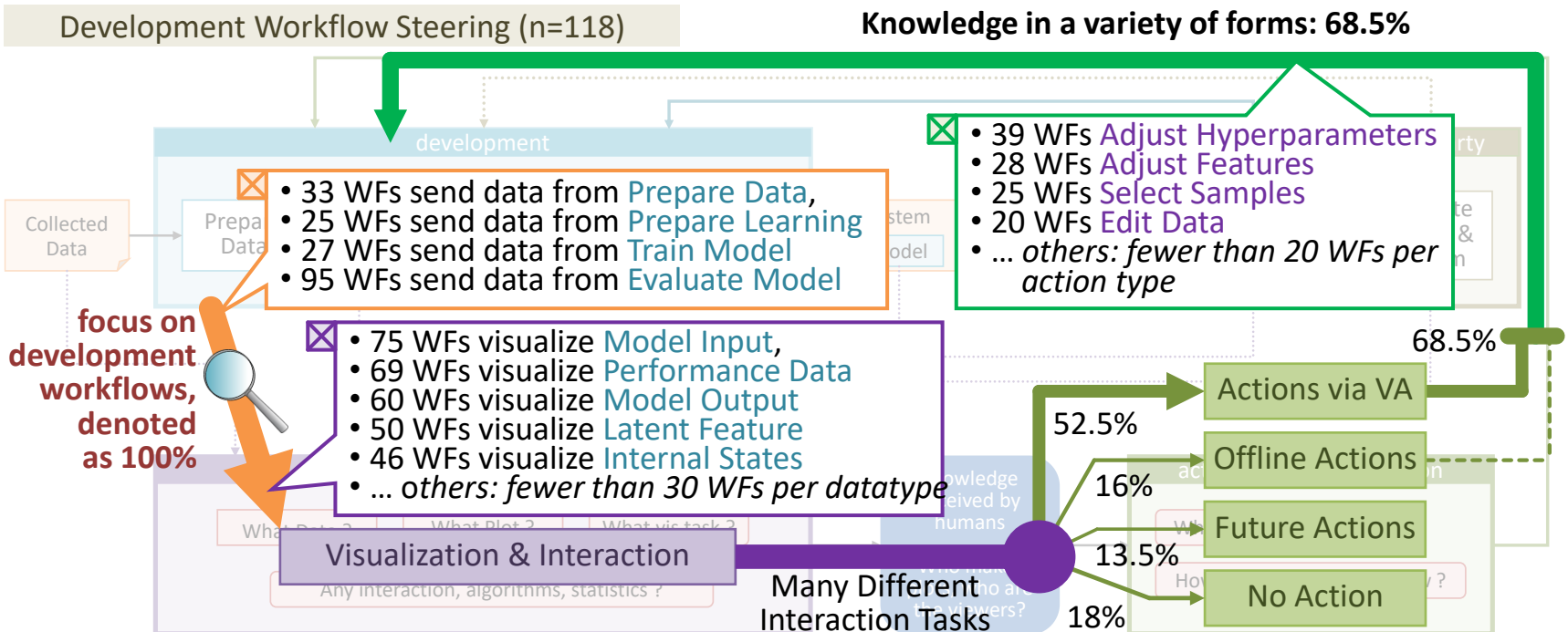
Development Workflow Steering (n=118)
Knowledge in a variety of forms: 68.5%
development
Collected Data
Prepa Data
• 33 WFs send data from Prepare Data,
• 25 WFs send data from Prepare Learning
• 27 WFs send data from Train Model
• 95 WFs send data from Evaluate Model
• 39 WFs Adjust Hyperparameters
• 28 WFs Adjust Features
• 25 WFs Select Samples
• 20 WFs Edit Data
• … others: fewer than 20 WFs per action type
focus on development workflows, denoted as 100%
• 75 WFs visualize Model Input,
• 69 WFs visualize Performance Data
• 60 WFs visualize Model Output
• 50 WFs visualize Latent Feature
• 46 WFs visualize Internal States
• … others: fewer than 30 WFs per datatype
68.5%
Actions via VA
52.5%
Offline Actions
16%
Future Actions
13.5%
No Action
18%
Visualization & Interaction
Many Different Interaction Tasks
Any interaction, algorithms, statistics ?